



SpaceX Falcon 9 First Stage Landing Prediction

APPLIED DATA SCIENCE CAPSTONE
PROJECT

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September 2026

Project Overview

Goal: Predict whether a Falcon 9 booster lands safely or is lost.

Why it matters: SpaceX charges ~\$62M per launch thanks to booster reuse, while non-reusable options run closer to \$165M.

Workflow: Data collection (API & Scraping) -> SQL analysis -> EDA -> Interactive mapping & dashboarding -> ML modeling.

Final Takeaway: Reached **83.33% accuracy** on unseen test data across optimized models.

The Business Case

Cost Difference: Reusing the first stage cuts the largest single cost driver in orbital launches.

Competitive Edge: Competitors need to estimate SpaceX's margins accurately to price their own commercial satellite bids.

Target Variable: Binary outcome (Class = 1 for landing success, 0 for crash/loss).

Data Ingestion (API & Scraping)

SpaceX API: Pulled 90 historical launches with payload, orbit, core, and site details.

Web Scraping: Scraped Wikipedia launch records with BeautifulSoup to cross-check records.

Result: A clean baseline dataset tracking launches from early testing through high-cadence operations.

```
# Load pre-extracted SpaceX dataset directly from IBM Cloud Storage
dataset_url = 'https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-QS0321EN-SkillsNetwork/datasets/dataset_part_1.csv'
data_falcon9 = pd.read_csv(dataset_url)

# Display shape to confirm 90 rows and 17 columns
print("Data shape:", data_falcon9.shape)
data_falcon9.head()
```

Data shape: (90, 17)

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857

Data Cleaning & Wrangling

Handled null values in PayloadMass and landing pad IDs.

Mapped outcomes: Successful recoveries (ocean splashdowns, drone ship landings, ground pads) were labeled 1; destroyed boosters and missed landings labeled 0.

Filtered down to high-impact variables for subsequent modeling.

```
data_falcon9.isnull().sum()
```

FlightNumber	0
Date	0
BoosterVersion	0
PayloadMass	5
Orbit	0
LaunchSite	0
Outcome	0
Flights	0
GridFins	0
Reused	0
Legs	0
LandingPad	26
Block	0
ReusedCount	0
Serial	0
Longitude	0
Latitude	0
dtype:	int64

SQL Exploratory Analysis

Loaded the data into SQLite
(my_data1.db) for rapid querying.

Key Findings:

Tracked launch counts across CCAFS, KSC, and VAFB.

Identified the first successful landing milestone.

Filtered boosters handling maximum payload capacities and reviewed the history of drone ship vs. ground pad recoveries.

Task 7

List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome", COUNT(*) AS Total_Count FROM SPACEXTABLE GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

Done.

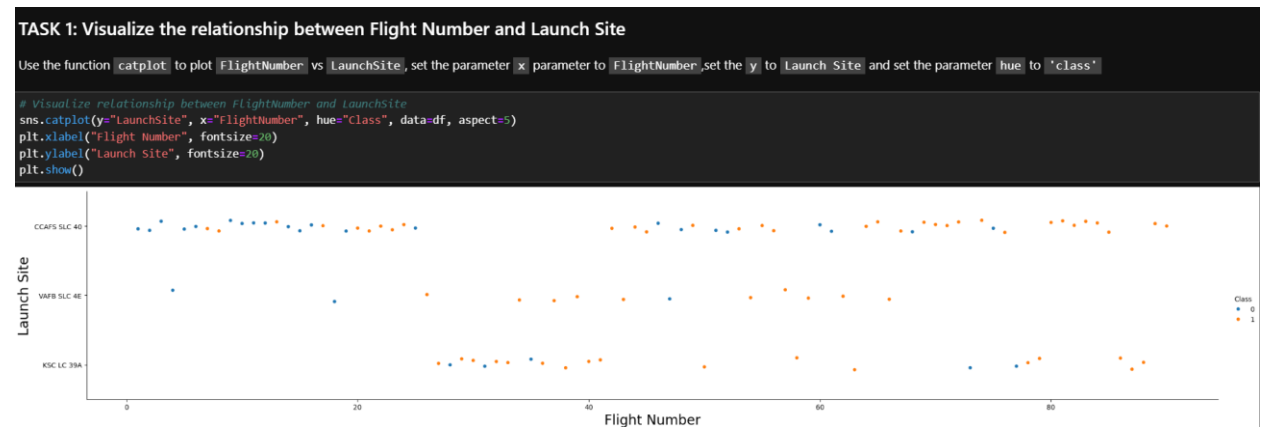
Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

EDA - Flight Number vs. Launch Site

CCAFS LC-40 handled the bulk of initial development flights.

KSC LC-39A and VAFB SLC-4E took on steady volume as launch rates grew.

Success rates visibly improve in higher flight numbers across every site.

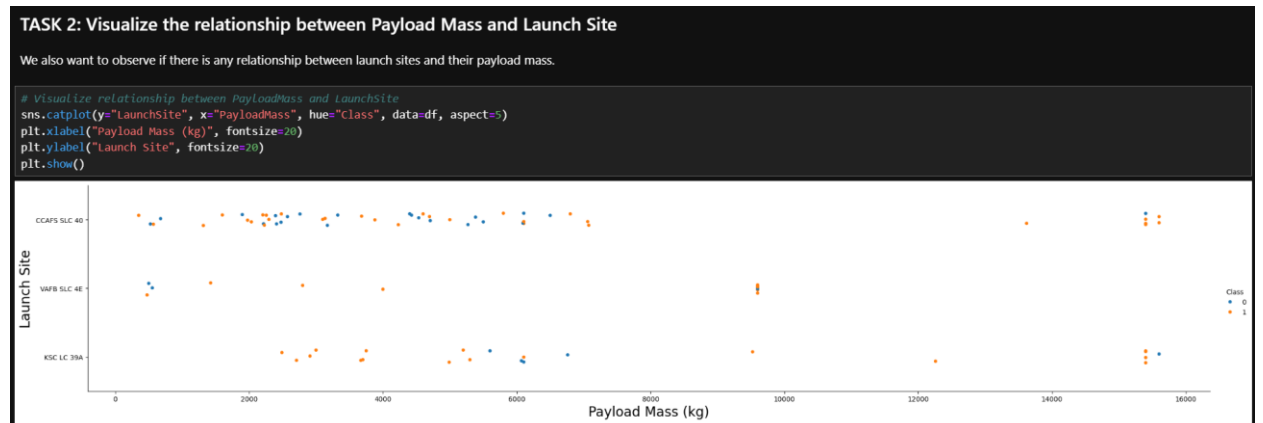


EDA - Payload Mass vs. Launch Site

KSC LC-39A and CCAFS LC-40 handle the heaviest payloads (up to ~15,000 kg).

VAFB SLC-4E focuses on lighter payloads headed to polar orbits.

Falcon 9 successfully recovered boosters across both low-mass and heavy-mass missions.



Success Rate by Orbit Type

Perfect Records: 100% success rate on ES-L1, GEO, HEO, and SSO profiles.

Low Earth Orbit (LEO & ISS): High recovery rates due to modest fuel requirements for the return burn.

GTO Missions: Lower success rates—high return velocity leaves less fuel margin and causes greater thermal stress.

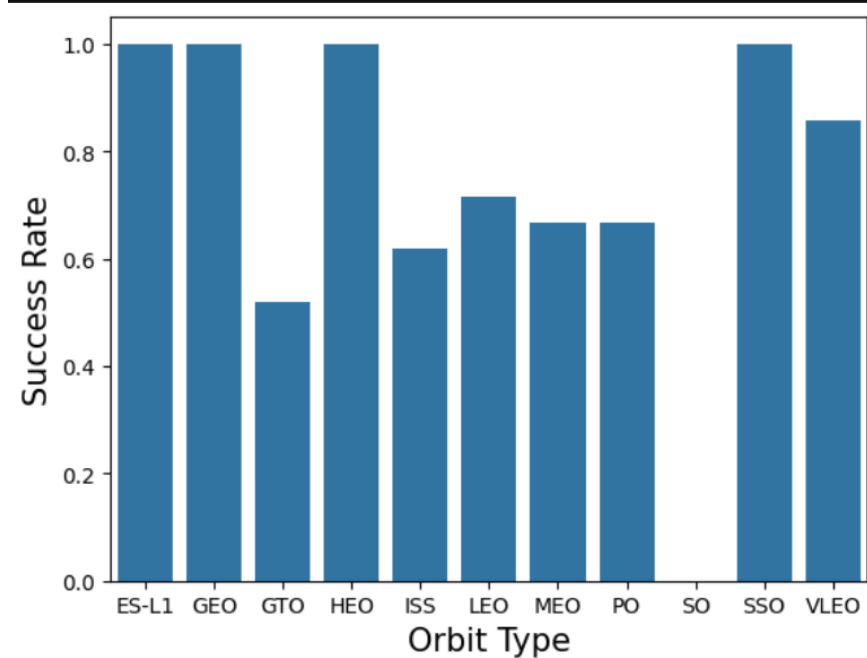
TASK 3: Visualize the relationship between success rate of each orbit type

Next, we want to visually check if there are any relationship between success rate and orbit type.

Let's create a `bar chart` for the success rate of each orbit

```
# Calculate success rate for each orbit type
orbit_success = df.groupby('Orbit')['Class'].mean().reset_index()

# Plot success rate by orbit type
sns.barplot(x="Orbit", y="Class", data=orbit_success)
plt.xlabel("Orbit Type", fontsize=15)
plt.ylabel("Success Rate", fontsize=15)
plt.show()
```



Yearly Success Trend

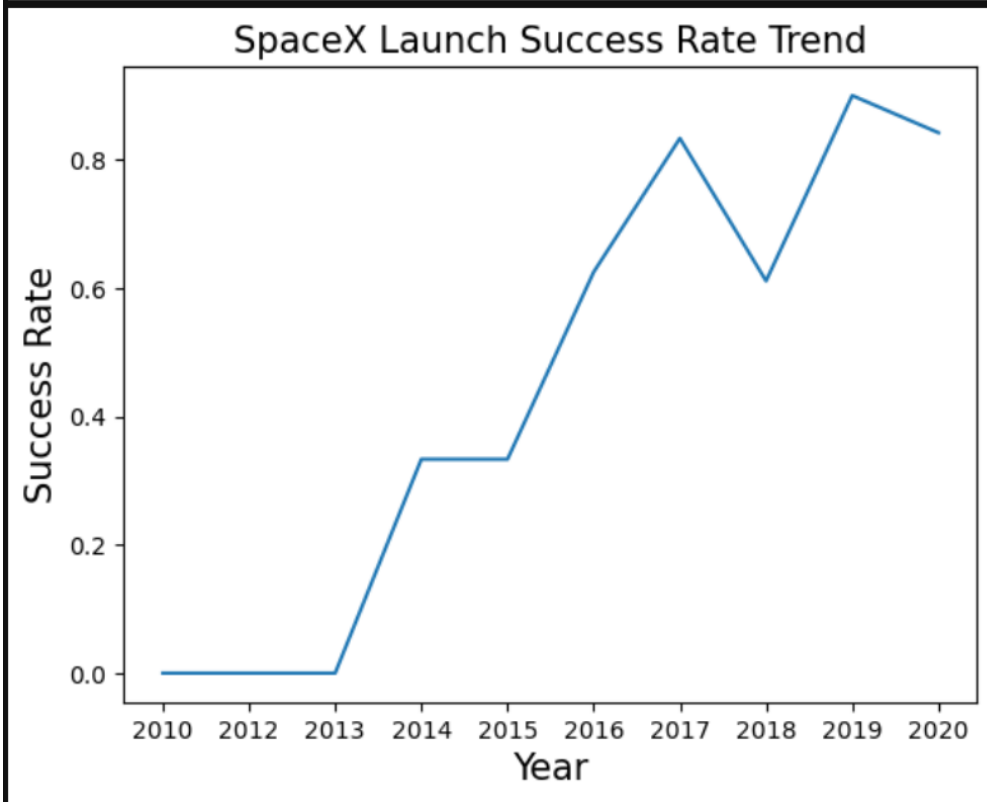
2010–2013: 0% recovery during early experimental phase.

Post-2015: Rapid improvement once grid fins, landing legs, and drone ship procedures stabilized.

2020: Recovery success climbed above 80%.

```
# Calculate yearly average success rate
yearly_success = df.groupby('Date')['Class'].mean().reset_index()

# Plot launch success yearly trend
sns.lineplot(x="Date", y="Class", data=yearly_success)
plt.xlabel("Year", fontsize=15)
plt.ylabel("Success Rate", fontsize=15)
plt.title("SpaceX Launch Success Rate Trend", fontsize=15)
plt.show()
```

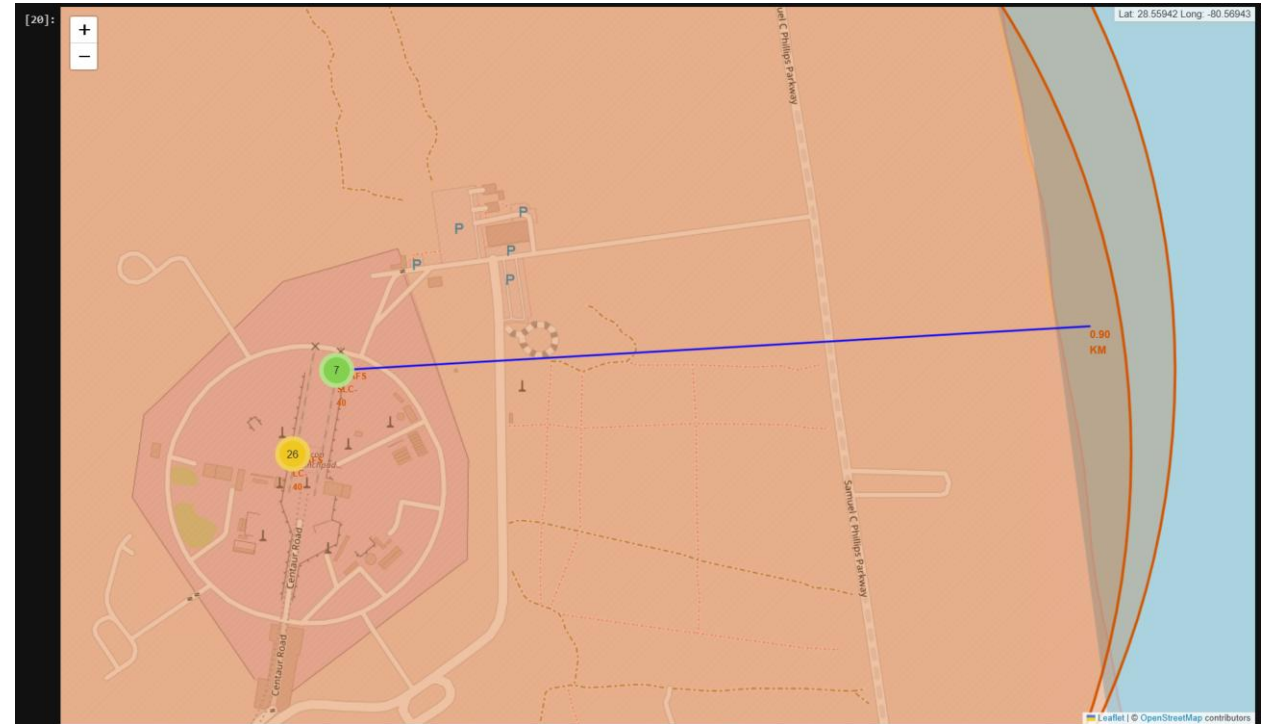


Geospatial Analysis (Folium)

Coastline Positioning: All pads sit directly on coastlines to keep abort and launch trajectories over the ocean.

Transport Logistics: Built near railways and highways for heavy booster transport while maintaining safety buffers from cities.

Recovery Zones: Offshore drone ship coordinates align closely with orbital launch azimuths.



Interactive Dashboard (Plotly Dash)

Built an interactive dashboard to test site and payload filters in real time.

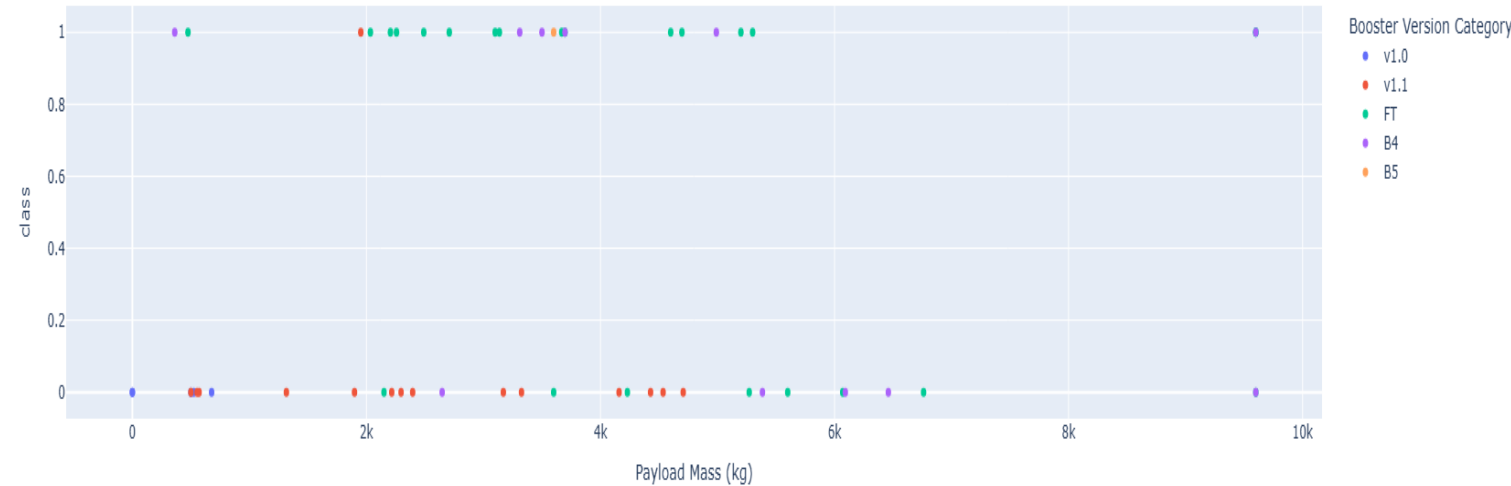
KSC LC-39A maintained the highest overall landing success rate.

Boosters carrying mid-range payloads (2,000–4,000 kg) and heavy loads (9,000+ kg) showed strong recovery rates.

Total Success Launches By Site



Correlation between Payload and Success for all Sites



Feature Preparation

One-hot encoded categorical columns (Orbit, LaunchSite, LandingPad, Serial), expanding feature space to 83 inputs.

Converted all columns to float64 for numerical consistency.

Scaled features with StandardScaler.

Split into 80% train (72 samples) and 20% test (18 samples) with a fixed random state.

TASK 8: Cast all numeric columns to float64

Now that our `features_one_hot` dataframe only contains numbers, cast the entire dataframe to variable type `float64`

```
# Cast all numeric columns to float64
features_one_hot = features_one_hot.astype('float64')

# Export to CSV for subsequent Machine Learning Lab
features_one_hot.to_csv('dataset_part_3.csv', index=False)

# Check the total number of columns
features_one_hot.shape

(90, 80)
```

Modeling Setup

Evaluated 4 core algorithms: Logistic Regression, SVM, Decision Tree, and KNN.

Tuned hyperparameters using GridSearchCV with 10-fold cross-validation on the training split.

Training cross-validation accuracy landed between 84% and 86% across all four setups.

Decision Tree Results

Best Parameters: criterion='entropy',
max_depth=14,
max_features='sqrt',
min_samples_leaf=2,
splitter='random'.

CV Score: 85.89% training accuracy.

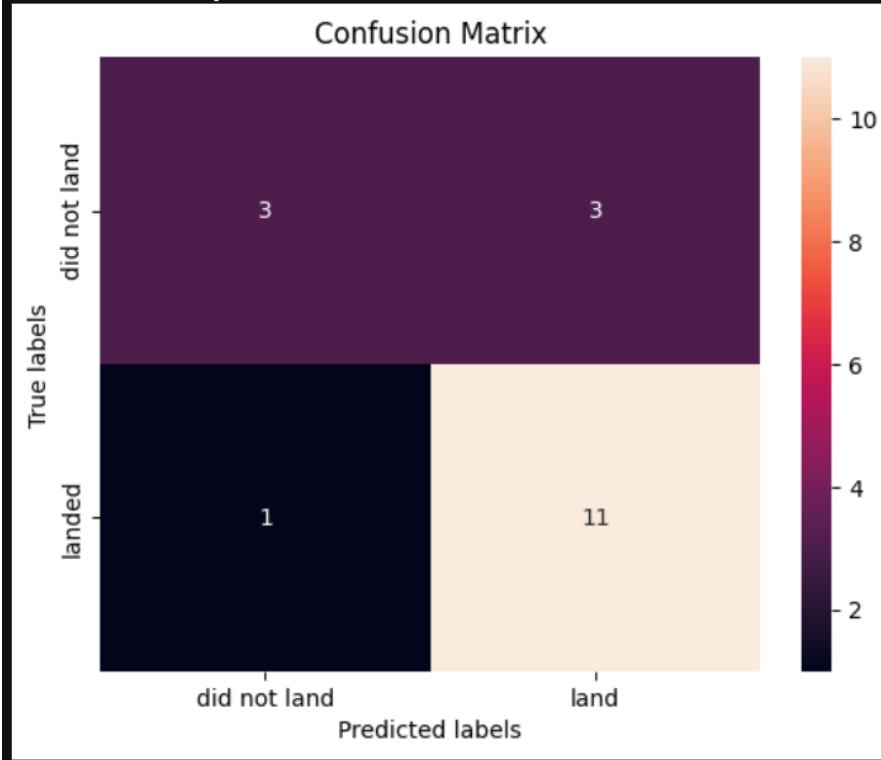
Confusion Matrix: Correctly
identified landing successes with
few false negatives on the test set.

Calculate the accuracy of tree_cv on the test data using the method `score` :

```
# Evaluate Decision Tree test accuracy
tree_test_score = tree_cv.score(X_test, Y_test)
print("Test set accuracy: ", tree_test_score)

# Predict test labels and plot confusion matrix
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test, yhat)
```

Test set accuracy: 0.7777777777777778



Model Comparison

Every tuned classifier reached **83.33% accuracy** on the 18 test samples (15 correct out of 18).

Given the small test set, all four models performed equivalently; Logistic Regression and SVM offer the most stable decision boundaries.

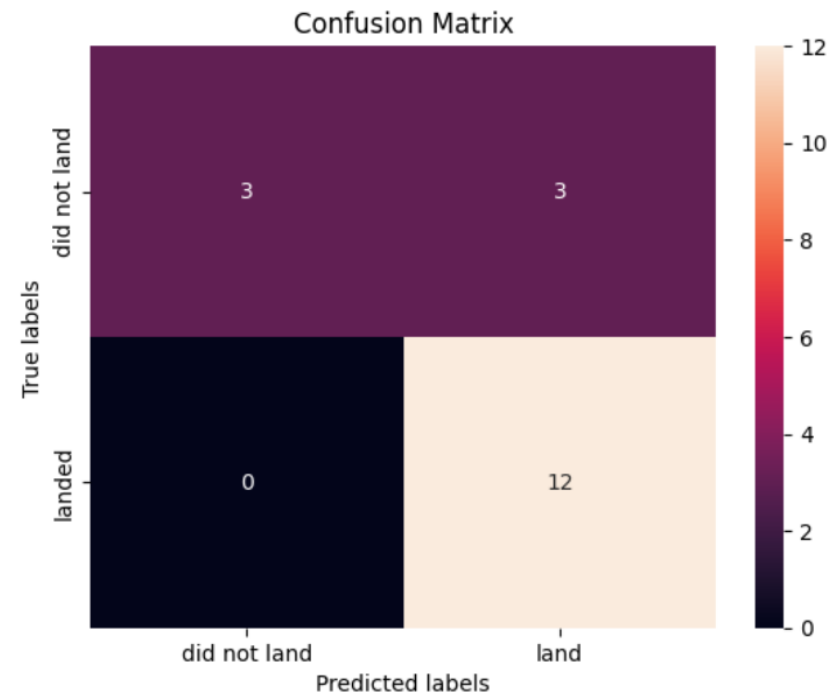
TASK 12

Find the method performs best:

```
# Evaluate KNN test accuracy
knn_test_score = knn_cv.score(X_test, Y_test)
print("Test set accuracy: ", knn_test_score)

# Predict test labels and plot confusion matrix
yhat = knn_cv.predict(X_test)
plot_confusion_matrix(Y_test, yhat)
```

Test set accuracy: 0.8333333333333334



Practical Takeaways

Experience matters: Recovery rates jumped significantly once landing hardware matured after 2015.

Orbit type is a primary risk factor: Heavy GTO missions leave tight fuel margins, increasing failure risk.

Launching from coastal sites with dedicated drone ships downrange is central to high recovery rates.

Strategic Recommendations for Competitors

Compete on LEO: Expect SpaceX to bid aggressively on LEO missions, where booster recovery is almost routine.

Target GTO: Missions requiring high-energy orbits present the highest risk of SpaceX expending a core—creating an opening for competitive bids.

Build Reusability: Long-term market parity requires developing sea-recovery assets and reusable first stages.

Summary & Repository

End-to-end analysis from raw data to functional ML models completed.

Full code, data, and dashboard files are organized on GitHub: github.com/onurtorun/IBM-Applied-Data-Science-Capstone

Open for questions.